Free APIs & Resources for Email Security Analysis

Executive Summary

This document provides a comprehensive catalog of **62 free and open-source APIs, tools, and resources** specifically curated for building a powerful email security analysis webapp. All resources listed are either completely free, have generous free tiers, or are open-source projects available on GitHub.

Quick Reference Guide

Best Free APIs (No Cost, High Value)

Service	Category	Free Tier	Best For
IP-API	IP Geolocation	45 req/min	IP location & proxy detection
Who-Dat	WHOIS	Unlimited	Domain registration lookup
EmailRep.io	Email Reputation	Rate limited	Email risk scoring
Google Safe Browsing	Phishing	10,000/day	URL safety checking
AlienVault OTX	Threat Intel	Unlimited	IoC and threat data
Verify Email API	Email Validation	Unlimited	Disposable email detection

Complete Resource Catalog

1. Email Header Analysis (3 Resources)

IPTrackerOnline Email Header Analyzer

- **URL**: https://www.iptrackeronline.com/email-header-analysis/
- **Type**: Free web tool
- **Features**: IP tracing, spam detection, SPF/DKIM/DMARC checks
- **Limitations**: Web interface only

Trustifi Email Analyzer

- **URL**: https://trustifi.com/email-analyzer/
- **Features**: Security protocol detection, delay identification
- **Cost**: Free

MXToolbox Email Header Analyzer

- **URL**: https://mxtoolbox.com/EmailHeaders.aspx
- **Features**: RFC822 parsing, human-readable format
- **Cost**: Free

2. IP Reputation Services (6 APIs)

AbuseIPDB ⭐

- **API**: REST API with free tier
- **URL**: https://www.abuseipdb.com/
- **Free Tier**: 1,000 checks/day
- **Features**: IP reputation scoring, abuse reports, confidence levels
- **API Key**: Required (free registration)
- **Best For**: Primary IP reputation checking

Antideo

- **URL**: https://www.antideo.com
- **Free Tier**: 10 API calls/hour
- **Features**: IP reputation, location, email validation, spam detection

IPQualityScore

- **URL**: https://www.ipqualityscore.com/
- **Free Tier**: 5,000 lookups/month
- **Features**: Proxy/VPN detection, fraud scoring, geolocation

IP2Location.io

- **URL**: https://www.ip2location.io/
- **Features**: Geolocation, proxy detection, abuse history

IP-API ⭐ *Highly Recommended*

- **API**: REST API (JSON/XML/CSV)
- **URL**: https://ip-api.com
- **Cost**: Free for non-commercial use
- **Limitations**: 45 requests/minute
- **No API Key**: Required
- **Features**: Geolocation, ISP, ASN, mobile, proxy, hosting detection

```
# Example Usage
import requests
```

```
def get_ip_info(ip_address):
    url = f'http://ip-api.com/json/{ip_address}'
    params = {
        'fields': 'status,country,city,isp,proxy,hosting'
    }
    response = requests.get(url, params=params)
    return response.json()
```

Spamhaus

- **URL**: https://www.spamhaus.org/
- **Type**: DNS-based blocklist
- **Features**: SBL, XBL, PBL, DBL lists
- **Cost**: Free for low-volume use

3. Phishing Detection (5 APIs)

Google Safe Browsing API ⭐

- **API**: REST API
- **URL**: https://developers.google.com/safe-browsing/v4
- **Free Tier**: 10,000 queries/day
- **Features**: Malware, phishing, unwanted software detection
- **API Key**: Required (Google Cloud)
- **Libraries**: Python, Go, Node.js available

PhishTank

- **API**: REST API
- **URL**: https://www.phishtank.com/api_info.php
- **Features**: Community-sourced phishing verification
- **API Key**: Required (free)

OpenPhish

- **Type**: Feed-based
- **URL**: https://openphish.com/
- **Free Version**: 12-hour update feed
- **Features**: Phishing URLs with timestamps

CheckPhish

- **URL**: https://developers.checkphish.ai/
- **Features**: AI-powered real-time phishing detection

- **Technology**: Deep learning, computer vision, NLP

Phishing Initiative

- **API**: REST API by CERT Orange Cyberdefense
- **URL**: https://phishing-initiative.eu/
- **GitHub**: https://github.com/CIRCL/PyEUPI
- **Cost**: Free

4. URL/Domain Reputation (4 APIs)

VirusTotal ⭐

- **API**: REST API
- **URL**: https://developers.virustotal.com/
- **Free Tier**: 4 requests/min, 500/day, 15.5K/month
- **Features**: 70+ AV engines scanning
- **Scans**: Files, URLs, domains, IP addresses
- **API Key**: Required

```
# VirusTotal Example
import requests

def scan_url_virustotal(url, api_key):
    vt_url = 'https://www.virustotal.com/api/v3/urls'
    headers = {'x-apikey': api_key}
    data = {'url': url}
    response = requests.post(vt_url, headers=headers, data=data)
    return response.json()
```

URLScan.io ⭐

- **API**: REST API
- **URL**: https://urlscan.io/docs/api/
- **Features**: Website screenshots, DOM analysis, network traffic capture
- **Free Tier**: Available
- **Best For**: Detailed URL analysis

IBM X-Force Exchange

- **URL**: https://api.xforce.ibmcloud.com/doc/
- **Features**: URL/IP/hash reputation, malware analysis
- **Free Tier**: Available

APIVoid URL Reputation

- **URL**: https://www.apivoid.com/api/url-reputation/
- **Features**: 50+ security checks, risk scoring
- **Pricing**: Credit-based (5 credits per call)

5. Email Validation (6 APIs)

EmailRep.io ⭐ *Open Source*

- **API**: REST API
- **URL**: https://emailrep.io
- **Cost**: Free with rate limits
- **Features**: Email reputation scoring, breach detection, social profiles
- **GitHub**: https://github.com/sublime-security/emailrep.io
- **No API Key**: Required for basic usage

```
# EmailRep Example
def check_email_reputation(email):
    url = f'https://emailrep.io/{email}'
    response = requests.get(url)
    return response.json()
```

Verify Email ⭐ *Fully Free*

- **API**: REST API
- **URL**: https://verifyemail.vercel.app/
- **Cost**: Completely free
- **Features**: Disposable/public provider detection
- **GitHub**: https://github.com/technikhil314/verify-email

Verifalia

- **URL**: https://verifalia.com/
- **Features**: Syntax validation, SMTP check, disposable detection
- **Free Tier**: Limited credits
- **SDKs**: .NET, Java, Node.js, Go, Ruby, PHP

Mailboxlayer

- **URL**: https://mailboxlayer.com
- **Free Tier**: 250 requests/month
- **Features**: SMTP validation, typo check

Maileroo

- **URL**: https://maileroo.com/free-email-verification-api

- **Free Tier**: 250 verifications/month
- **Features**: SMTP check, disposable detection, catch-all

Abstract Email Validation

- **URL**: https://www.abstractapi.com/
- **Free Tier**: 100 requests/month
- **Features**: Typo detection, SMTP verification

6. DNS Lookup Services (4 APIs)

HackerTarget DNS Lookup ⭐

- **API**: REST API (plain text/JSON)
- **URL**: https://hackertarget.com/dns-lookup/
- **Free Tier**: 50 queries/day
- **Features**: A, AAAA, MX, NS, CNAME, TXT, PTR, SOA records

```
# DNS Lookup Example
def dns_lookup(domain):
    url = f'https://api.hackertarget.com/dnslookup/?q={domain}'
    response = requests.get(url)
    return response.text
```

API Ninjas DNS Lookup

- **URL**: https://api-ninjas.com/api/dnslookup
- **Features**: A, AAAA, NS, PTR, SOA, CAA records
- **Note**: Premium required for TXT/MX records

DNSDumpster

- **Type**: Web tool
- **URL**: https://dnsdumpster.com
- **Features**: DNS reconnaissance, subdomain discovery
- **Cost**: Free

ZDNS ⭐ *Open Source*

- **Type**: CLI tool / Go library
- **GitHub**: https://github.com/zmap/zdns
- **Features**: High-speed DNS resolution
- **Best For**: Bulk DNS queries

7. SPF/DKIM/DMARC Validation (4 Tools)

DMARCLY

- **URL**: https://dmarcly.com
- **Features**: DMARC monitoring, SPF/DKIM checking
- **Free Tier**: Available

EasyDMARC Checker

- **URL**: https://easydmarc.com/tools/dmarc-lookup
- **Features**: DMARC record checking, SPF flattening
- **Trial**: 14-day free trial

Red Sift Investigate

- **URL**: https://redsift.com/tools/investigate
- **Type**: Dynamic checker (send test email)
- **Features**: SPF/DKIM/DMARC alignment, FCrDNS, TLS
- **Cost**: Free

MXToolbox DMARC Tools

- **URL**: https://mxtoolbox.com/dmarc/
- **Features**: Record checker, generator, XML analyzer
- **Cost**: Free tools

8. WHOIS Lookup (3 APIs)

Who-Dat ⭐ *Fully Open Source*

- **API**: REST API
- **URL**: https://who-dat.as93.net
- **Cost**: Completely free
- **No API Key**: Required
- **GitHub**: https://github.com/Lissy93/who-dat
- **Features**: WHOIS/RDAP lookup, no-CORS

```
# Who-Dat Example
def whois_lookup(domain):
    url = f'https://who-dat.as93.net/{domain}'
    response = requests.get(url)
    return response.json()
```

HackerTarget WHOIS

- **URL**: https://hackertarget.com/whois-lookup/
- **Free Tier**: 5 queries/day
- **Features**: Domain and IP WHOIS

Whoxy

- **URL**: https://www.whoxy.com
- **Pricing**: $2 per 1,000 queries (paid)
- **Features**: 2,026 TLDs supported, parsed data

9. Blacklist Checking (5 Services)

MXToolbox Blacklist Check

- **URL**: https://mxtoolbox.com/blacklists.aspx
- **Features**: Check against 100+ DNSBL
- **Cost**: Free

Blacklist Master

- **API**: REST API
- **URL**: https://www.blacklistmaster.com/apiv1
- **Features**: IP/domain blacklist checking, historical data
- **API Key**: Required

HetrixTools Blacklist API

- **URL**: https://docs.hetrixtools.com/blacklist-check-api/
- **Type**: Credit-based API
- **Features**: Real-time blacklist checking

RBLTracker

- **URL**: https://rbltracker.com
- **Features**: Real-time monitoring, REST API
- **Free Tier**: Limited monitoring

Blacklist Checker

- **URL**: https://blacklistchecker.com
- **Features**: 100+ blacklist checks
- **API**: Paid subscription for API access

10. Threat Intelligence Feeds (6 Sources)

AlienVault OTX ⭐ *Highly Recommended*

- **API**: REST API
- **URL**: https://otx.alienvault.com/api
- **Cost**: Completely free
- **Features**: IoCs, malware analysis, threat pulses
- **Python SDK**: Available
- **Community**: Large active community

Blocklist.de

- **Type**: Feed-based
- **URL**: https://www.blocklist.de/
- **Features**: SSH, FTP, email, webserver attack IPs
- **Cost**: Free

Emerging Threats

- **Type**: Feed-based (Proofpoint)
- **URL**: https://rules.emergingthreats.net/
- **Free Version**: Open source
- **Features**: 40+ threat categories, confidence scores

hpHosts

- **Type**: Hosts database
- **Features**: Community-managed malicious hosts
- **Note**: Last updated 2019, but still reliable

CrowdSec

- **API**: REST API
- **URL**: https://www.crowdsec.net/
- **Free Tier**: 50 queries/day
- **Features**: 25M+ malicious IPs, contextual data
- **GitHub**: https://github.com/crowdsecurity

Shadowserver

- **Type**: Email/API reports
- **URL**: https://www.shadowserver.org/
- **Cost**: Free
- **Features**: Botnet, exposed services, malware reports

11. Malware Analysis Sandboxes (5 Services)

Hybrid Analysis ⭐

- **API**: REST API
- **URL**: https://www.hybrid-analysis.com/
- **Free Tier**: Available with rate limits
- **Features**: File/URL analysis, behavior reports

ANY.RUN

- **URL**: https://any.run/
- **Type**: Interactive sandbox
- **Free Tier**: Limited submissions
- **Features**: Real-time interaction, screenshots

Joe Sandbox

- **URL**: https://www.joesecurity.org/
- **API**: Yes
- **Free Tier**: Available
- **Features**: Deep malware analysis

Cuckoo Sandbox ⭐ *Open Source*

- **Type**: Self-hosted
- **GitHub**: https://github.com/cuckoosandbox/cuckoo
- **Cost**: Free (open source)
- **Features**: Automated analysis, extensible, API

CAPE Sandbox ⭐ *Open Source*

- **Type**: Self-hosted
- **URL**: https://capesandbox.com
- **GitHub**: https://github.com/kevoreilly/CAPEv2
- **Features**: Config extraction, YARA signatures, unpacking
- **Cost**: Free (open source)

12. SSL/TLS Validation (3 Services)

ZeroSSL

- **API**: REST API
- **URL**: https://zerossl.com/developer
- **Features**: Certificate management API
- **Requirements**: Pro plan for unlimited API

SSL For Free

- **URL**: https://www.sslforfree.com
- **Features**: Free 90-day SSL certificates
- **Provider**: Powered by ZeroSSL

Let's Encrypt ⭐

- **Protocol**: ACME
- **URL**: https://letsencrypt.org/
- **Cost**: Completely free
- **Features**: Automated certificate issuance/renewal
- **Tools**: Certbot, acme.sh, many ACME clients

13. OSINT Tools (3 Platforms)

Maltego Community Edition

- **URL**: https://www.maltego.com/
- **Type**: Desktop application
- **Free Tier**: Limited transforms, 24 results per transform
- **Features**: Visual link analysis, 100+ data sources

SpiderFoot *Open Source*

- **GitHub**: https://github.com/smicallef/spiderfoot
- **Features**: 200+ modules, threat intelligence automation
- **Cost**: Free (open source)
- **Note**: GitHub version outdated since acquisition

OSINT Framework

- **URL**: https://osintframework.com
- **Type**: Tool directory
- **GitHub**: https://github.com/lockfale/OSINT-Framework
- **Features**: Organized directory of OSINT resources

14. GitHub Open Source Projects (5 Projects)

Gophish ⭐

- **URL**: https://getgophish.com
- **GitHub**: https://github.com/gophish/gophish
- **Type**: Phishing framework
- **Features**: Campaign management, templates, REST API
- **Cost**: Free (open source)

DRAKVUF Sandbox

- **GitHub**: https://github.com/CERT-Polska/drakvuf-sandbox
- **Type**: Malware analysis sandbox
- **Features**: Agentless black-box analysis
- **Platforms**: Windows 7, Windows 10

Qu1cksc0pe

- **GitHub**: https://github.com/CYB3RMX/Qu1cksc0pe
- **Type**: All-in-one malware analysis tool
- **Features**: Static/dynamic analysis, multiple file types
- **Platforms**: Windows, Linux, Android, macOS

Malware Jail

- **GitHub**: https://github.com/HynekPetrak/malware-jail
- **Type**: JavaScript malware sandbox
- **Platform**: Node.js
- **Features**: Deobfuscation, payload extraction

OSINT APIs Collection ⭐

- **GitHub**: https://github.com/cipher387/API-s-for-OSINT
- **Type**: Comprehensive API directory
- **Features**: Curated list of OSINT APIs
- **Categories**: 20+ categories of APIs

🎯 Implementation Strategy

Tier 1: Essential (Implement First)

1. **IP-API** - IP geolocation and reputation
2. EmailRep.io - Email reputation checking
3. **Google Safe Browsing** - Phishing detection
4. **Who-Dat** - WHOIS lookups
5. **HackerTarget** - DNS queries
6. **Verify Email API** - Email validation

Tier 2: Enhanced Features

7. **AbuseIPDB** - Advanced IP reputation
8. **VirusTotal** - Multi-engine scanning
9. URLScan.io - URL analysis
10. **AlienVault OTX** - Threat intelligence
11. **MXToolbox** - Blacklist checking

Tier 3: Advanced Capabilities

12. **Hybrid Analysis** - Malware sandboxing
13. **CAPE Sandbox** - Self-hosted analysis
14. **Cuckoo Sandbox** - Self-hosted automation

Required Features (From Attachment)

Based on the `extra-features-needs-to-be-added.xlsx`:

Command	Feature	Implementation
`blacklist:`	IP/host reputation	AbuseIPDB, Spamhaus
`smtp:`	SMTP server test	Port 25 connection test
`mx:`	MX records	DNS query
`a:`	A record	DNS query
`spf:`	SPF validation	TXT record parse
`txt:`	TXT records	DNS query
`ptr:`	Reverse DNS	PTR lookup
`cname:`	CNAME lookup	DNS query
`whois:`	Domain info	Who-Dat API
`arin:`	IP block info	ARIN database

Command	Feature	Implementation
`soa:`	SOA record	DNS query
`tcp:`	TCP test	Socket connection
`http:`	HTTP test	HTTP request
`https:`	HTTPS test	HTTPS request
`ping:`	ICMP ping	Ping command
`trace:`	Traceroute	Traceroute
`dns:`	DNS check	DNS validation

Security Best Practices

API Key Management

- Store in environment variables or secure vault
- Never commit to version control
- Rotate keys regularly
- Use separate keys for dev/prod

Input Validation

- Sanitize all email headers
- Validate IPs, domains, URLs
- Prevent injection attacks
- Implement rate limiting

Data Privacy

- Don't log sensitive email content
- Anonymize stored data
- Comply with GDPR/privacy regulations
- Implement data retention policies

Error Handling

- Don't expose API keys in errors
- Provide user-friendly messages
- Log detailed errors securely
- Implement fallback mechanisms

API Rate Limits Summary

Service	Free Tier Limit	Notes
VirusTotal	4/min, 500/day	Most restrictive
Google Safe Browsing	10,000/day	Very generous
IP-API	45/minute	No registration
AbuseIPDB	1,000/day	Registration required
HackerTarget	50/day	Various endpoints
EmailRep.io	Rate limited	No hard limit stated
AlienVault OTX	Unlimited	Registration required
Who-Dat	Unlimited	Open source

Best Practices for Integration

1. Asynchronous Operations

```
import asyncio
import aiohttp

async def check_multiple_sources(ip_address):
    tasks = [
        check_abuseipdb(ip_address),
        check_ip_api(ip_address),
        check_spamhaus(ip_address)
    ]
    results = await asyncio.gather(*tasks)
    return aggregate_results(results)
```

2. Caching Strategy

- Cache IP reputation for 6 hours
- Cache domain reputation for 12 hours
- Cache WHOIS data for 24 hours
- Use Redis or similar for caching

3. Error Handling with Fallbacks

```
async def check_with_fallback(primary, fallback, *args):
    try:
        result = await primary(*args)
        if result.get('success'):
            return result
    except Exception:
```

```
            logging.warning("Primary failed, using fallback")

        return await fallback(*args)
```

4. Rate Limit Management

- Implement token bucket algorithm
- Queue requests when approaching limits
- Provide user feedback on quota usage
- Distribute load across multiple APIs

Additional Resources

Documentation

- Email Security Best Practices (OWASP)
- Threat Intelligence Sharing (STIX/TAXII)
- DNS Security (DNSSEC)

GitHub Collections

- Awesome OSINT
- Awesome Threat Intelligence
- Awesome Malware Analysis

Learning Resources

- SANS Internet Storm Center
- Recorded Future Blog
- Krebs on Security

Conclusion

This comprehensive guide provides **62 free and open-source resources** for building a robust email security analysis platform. The resources are categorized, prioritized, and come with implementation guidance.

Key Takeaways:

✓ All resources are free or have generous free tiers
✓ Open-source alternatives available for self-hosting
✓ Multiple fallback options for reliability

✅ Implementation examples provided
✅ Security best practices included

Next Steps:

1. Review the Tier 1 essential APIs
2. Register for free API keys
3. Implement modular service architecture
4. Add caching and rate limiting
5. Deploy with monitoring and logging

Remember: Always respect API rate limits, implement proper error handling, and prioritize user privacy and data security.

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All Resources: 100% Free or Open Source